DISTRIBUTED SYSTEM AND METHOD FOR COVERT COMMUNICATION VIA SATELLITE-OBSERVABLE, GEO-SPATIALLY LOCATED VISUAL ENCODERS

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Title: DISTRIBUTED SYSTEM AND METHOD FOR COVERT COMMUNICATION VIA SATELLITE-OBSERVABLE, GEO-SPATIALLY LOCATED VISUAL ENCODERS

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Abstract

A system and method for secure, untraceable communication and operational coordination of a distributed network of agents is disclosed. The system architecture utilizes a novel, two-tiered sharding protocol for dynamic task allocation to a plurality of mobile assets, such as vehicles. An "object sharding" function first partitions assets into groups based on physical features, after which a time-variant "function sharding" protocol dynamically assigns operational roles to said groups, creating a resilient and unpredictable operational pattern. Communication, including the transmission of status messages and encryption key seeds, is achieved without reliance on the conventional radio frequency (RF) spectrum. Instead, the system uses satellite-observable "visual encoders"—arrangements of large physical objects

of specific colors, shapes, and orientations—to transmit data. A highly secure, two-factor method for transmitting encryption key seeds is also disclosed, wherein a first seed component is transmitted via a satellite-observable visual encoder at a first geographical location, and a second component is transmitted via a local, non-RF channel at a second location. The first location is mathematically derived from the second by applying a predetermined coordinate transformation, such as negating the longitude, thus obviating the need for a central database of keying locations and providing tiered security.

Background of the Invention

Clandestine and intelligence networks have historically relied on radio frequency (RF) communications for command, control, and data exfiltration. However, RF transmissions are inherently vulnerable to detection and interception by signals intelligence (SIGINT) operations. As SIGINT technologies have become more sophisticated, the risk of compromise for any network relying on traditional electronic emissions has grown exponentially. This has created a need for novel communication methods that are difficult to detect, intercept, or attribute, preferably operating outside the regulated and heavily monitored RF spectrum.

Furthermore, managing a large, distributed network of field assets presents significant logistical challenges. Static assignment of roles and responsibilities can create predictable patterns that are easily identified and exploited by adversaries. There is a need for a command-and-control architecture that allows for the dynamic and flexible tasking of assets in a manner that appears random and is resistant to pattern analysis.

Finally, the secure distribution of cryptographic keys remains a fundamental challenge for any secure communication system. Traditional key exchange protocols often require an electronic transaction that can be detected. A method for securely distributing keying material that does not create a detectable electronic or RF-based event is therefore highly desirable for secure network operations. The present invention addresses these limitations by providing an integrated system for dynamic asset management and secure, untraceable communication using satellite-observable visual information.

Summary of the Invention

The present invention discloses a multi-layered system for managing a distributed network of assets and conducting secure communications without using traditional RF transmissions.

First, the invention provides a novel system architecture for asset management based on a two-tiered sharding protocol. A plurality of assets (e.g., vehicles) are first grouped

into "object shards" based on deterministic, observable features (e.g., color, vehicle type). A second, "function sharding" protocol then dynamically assigns operational roles (e.g., surveillance, listening, logistical support) to these shards based on a time-variant function (e.g., the day of the week or hour of the day). This dynamic, multi-layered sharding creates an operational posture that is highly flexible and difficult for an outside observer to predict.

Second, the invention provides a method for transmitting data, such as status messages or encryption key seeds, by creating "visual encoders" on the ground that are observable by imaging satellites. These encoders are formed by arranging large, physical objects of predetermined colors, shapes, and orientations into specific spatial patterns. Examples include arranging colored landscaping equipment on an open trailer or parking asymmetrically shaped vehicles at specific angles on a semi-circular driveway to encode a vector or directional information. The satellite captures an image of this pattern, which is then decoded to retrieve the message.

Third, the invention provides a highly secure, two-factor method for transmitting encryption key seeds. A first portion of the seed is encoded and transmitted using a satellite-observable visual encoder at a first geographic location. A second portion is transmitted via a local, non-RF channel (e.g., Near-Field Communication) at a second, different geographic location. To reconstruct the key, an adversary must intercept both transmissions. Security is further enhanced by mathematically deriving the first location from the second (e.g., by negating one or both of the coordinate values), creating a tiered security system that does not require a vulnerable central database of keying locations. Methods for conducting such communications at night by creating "islands of light" are also disclosed.

Brief Description of the Drawings

FIG. 1 is a high-level diagram illustrating the overall system architecture, including satellite assets, ground agents, and targets.

FIG. 2 is a conceptual diagram illustrating the two-tiered "object sharding" and "function sharding" protocol for dynamic asset allocation.

FIG. 3 is a diagram illustrating a visual encoder for satellite communication, using colored objects on a trailer.

FIG. 4 is a diagram illustrating a method of encoding directionality using an asymmetrically-shaped vehicle on a semi-circular driveway.

FIG. 5 is a block diagram illustrating the two-factor method for encryption key seed transmission.

FIG. 6 is a world map illustrating the geo-relocation of a visual key seed from a location in

North America to a corresponding location in Asia by negating the longitude coordinate. **FIG. 7** is a diagram illustrating a night-time communication operation using an illuminated athletic field.

Detailed Description of the Invention

System Architecture: Two-Tiered Sharding

Referring to **FIG. 1**, the invention provides a system for managing a distributed network of mobile agents (101), often embodied as vehicles, for purposes of surveillance and communication. The system's operation is observed by one or more imaging satellites (102).

A core innovation of the system architecture is a two-tiered sharding protocol for dynamic task allocation, illustrated in **FIG. 2**. A heterogeneous fleet of assets (201) is first partitioned via an **Object Sharding** function (202). This function deterministically assigns each asset to a shard based on its static, observable physical features. For example, all gray sedans may be assigned to Shard A (203), while all white trucks with a specific character in their license plate may be assigned to Shard B (204).

Subsequently, a **Function Sharding** protocol (205) assigns a specific operational role or function to each shard. Crucially, this function is time-variant and depends on external variables, such as the day of the week or time of day. For example, a function 'F(Shard, Day)' might assign Shard A to a "Level 2 Surveillance" role (206) on Tuesdays, but a "Level 1 Logistics" role (207) on Saturdays. This dynamic, two-tiered assignment methodology creates an operational environment that is highly resistant to external pattern analysis, as the behavior of any given group of assets is not static. This implies that the hardware (e.g., sensors, communication devices) on the assets is largely homogeneous and capable of multiple roles, with the specific function being constrained by software at any given time.

Method for Covert Satellite Communication

Referring to **FIG. 3**, the system employs a method for communication that avoids the use of the traditional RF spectrum. Data is transmitted by creating "visual encoders" on the ground. A mobile agent arranges one or more physical objects (301) of predetermined color, shape, and size onto a platform, such as an open-air trailer (302). The spatial arrangement, orientation, color, and type of these objects form a specific pattern. An imaging satellite (102) captures an image of this pattern, which is then transmitted to a ground station for decoding, thus retrieving the encoded message.

A specific embodiment for encoding directionality or vector information is shown in **FIG. 4**. An agent's property may feature a semi-circular driveway (401). An asymmetrically-shaped vehicle (402), such as a sports car where the front is visually distinct and longer than the rear, is parked on the driveway. The precise angle or orientation (403) at which the vehicle is parked can be accurately determined from a satellite image. This angle serves to encode a value, status, or vector. The presence or absence of vehicles of different colors can encode additional states.

Method for Secure Key Seed Transmission

Referring to **FIG. 5**, the invention provides a highly secure two-factor method for distributing encryption keying material. An encryption seed is split into two components. The first component, the Visual Seed (501), is encoded and transmitted using the satellite-observable method described in FIG. 3. The second component, the Local Seed (502), is transmitted at a specific location via a local, non-RF channel, such as Near-Field Communication (NFC) from a passing vehicle (503) to a receiver (504) at a fixed premises. Both the Visual Seed and the Local Seed must be successfully received and combined (505) to reconstruct the full seed necessary to generate the final encryption key (506).

To further enhance security, the geographic locations of the two seed component transmissions are physically separated. As shown in **FIG. 6**, the system obviates the need for a vulnerable central database of keying locations by using a rule-based coordinate transformation. The location for the Visual Seed transmission (601) is mathematically derived from the location of the Local Seed transmission (602). In one embodiment, for a high-security asset, the longitude of the Local Seed location is negated (e.g., -95.8° becomes +95.8°), relocating the Visual Seed to the opposite side of the planet. Different transformations (e.g., negating latitude, or both) can be used to create a tiered security system based on pre-shared rules. The initial selection of secure premises (602) is strategically made such that its transformed counterpart (601) falls within a geographically shielded and difficult-to-observe area.

Night-Time Operations

The reliance on visible light presents a challenge for night-time operations. As shown in **FIG. 7**, this limitation is overcome by creating temporary "islands of light." An operation may be conducted under the pretext of a night-time event, such as an athletic game on a field with stadium-quality lighting (701). Agents (702), appearing as players, can then arrange colored objects (e.g., cones, equipment, uniforms) on the brightly-lit field (703). The intense illumination makes the colors and patterns of these objects clearly visible to an imaging

satellite (102), allowing the same communication methods to be employed at night.

Claims

What is claimed is:

1. A method for managing a distributed network of mobile assets, comprising the steps of:

    - Assigning each of a plurality of mobile assets to an object shard based on a first deterministic function of the asset's static physical features; and
    - Assigning an operational role to said object shard based on a second deterministic function, wherein said second function is time-variant and dependent on at least one external variable not derived from the asset's physical features.

2. The method of claim 1, wherein the static physical features include at least one of vehicle color, vehicle type, or a character in a license plate.

3. The method of claim 1, wherein the time-variant external variable is the day of the week or the time of day.

4. A method for covertly communicating information, comprising the steps of:

    - Arranging one or more non-luminous physical objects of predetermined color and shape into a spatial pattern within a geographic area;
    - Capturing a digital image of said spatial pattern with an earth-orbiting satellite; and
    - Decoding said digital image to extract an informational message represented by the arrangement, color, and shape of said physical objects.

5. The method of claim 4, wherein the physical objects are pieces of landscaping equipment on an open-air trailer.

6. The method of claim 4, wherein the physical object is an asymmetrically-shaped vehicle and the spatial pattern is the orientation of said vehicle on a semi-circular driveway.

7. A method for securely transmitting an encryption key seed, comprising the steps of:

    - Transmitting a first portion of the seed via a satellite-observable visual encoder at a first geographic location;

- Transmitting a second portion of the seed via a local, non-radio-frequency channel at a second geographic location, distinct from the first;
- Receiving both the first and second portions of the seed; and
- Combining the first and second portions to reconstruct the complete encryption key seed.

8. The method of claim 7, wherein the first geographic location is mathematically derived by applying a predetermined coordinate transformation to the coordinates of the second geographic location.

9. The method of claim 8, wherein the coordinate transformation comprises negating the latitude value, the longitude value, or both the latitude and longitude values of the coordinates of the second geographic location.

10. The method of claim 7, wherein the step of transmitting the first portion occurs at night within an artificially illuminated area.

Drawings

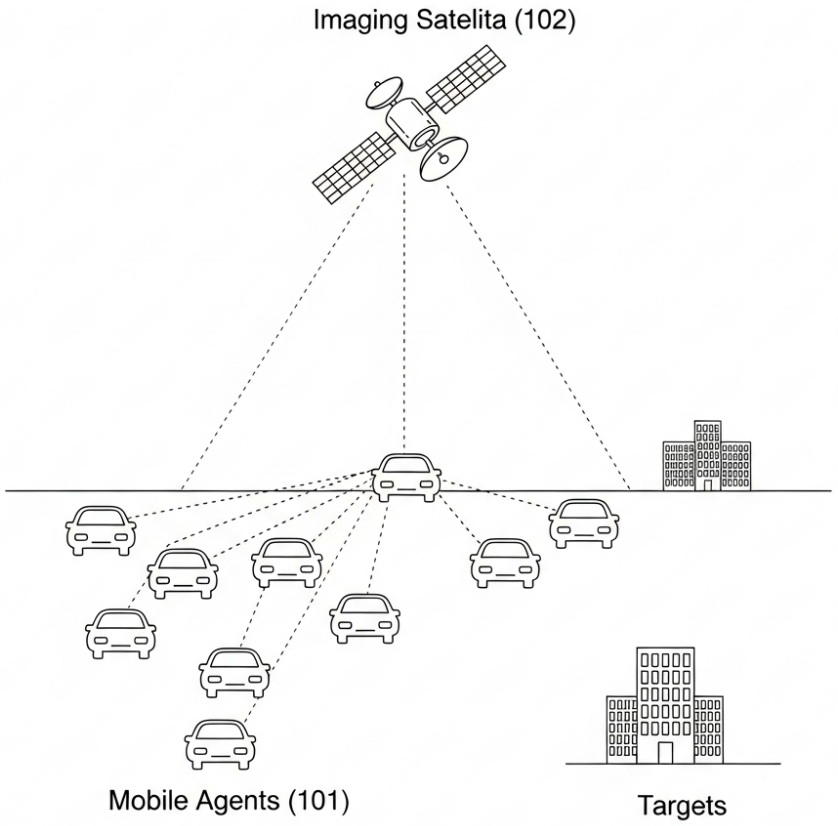


Figure 1: System architecture diagram.

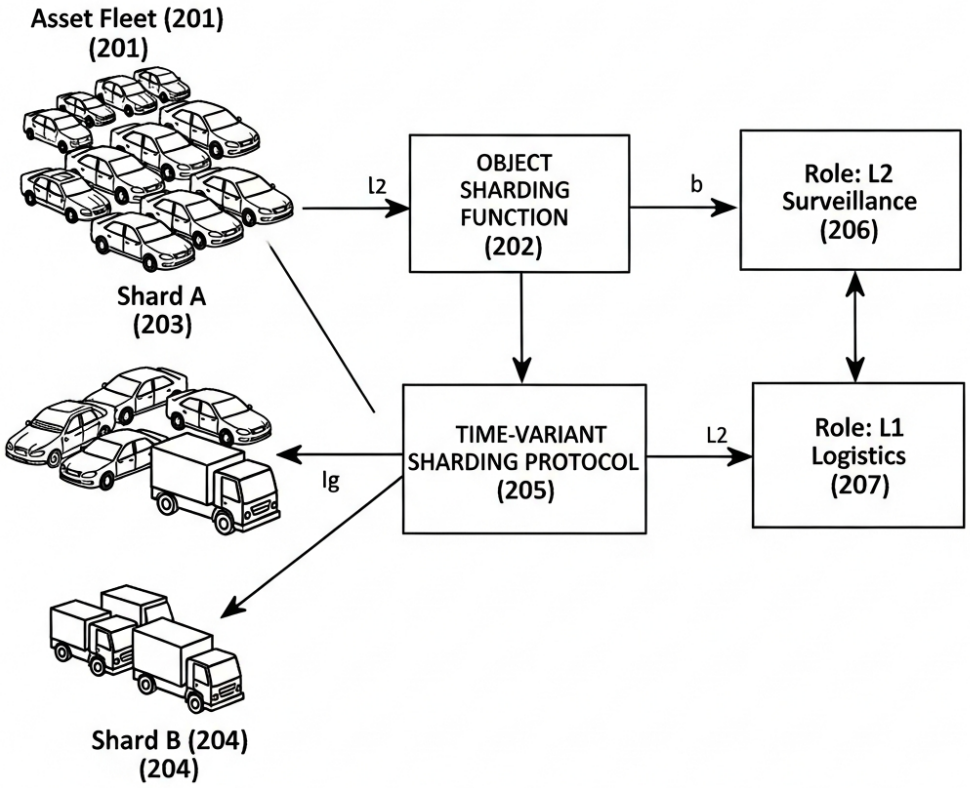


Figure 2: Two-tiered sharding protocol.

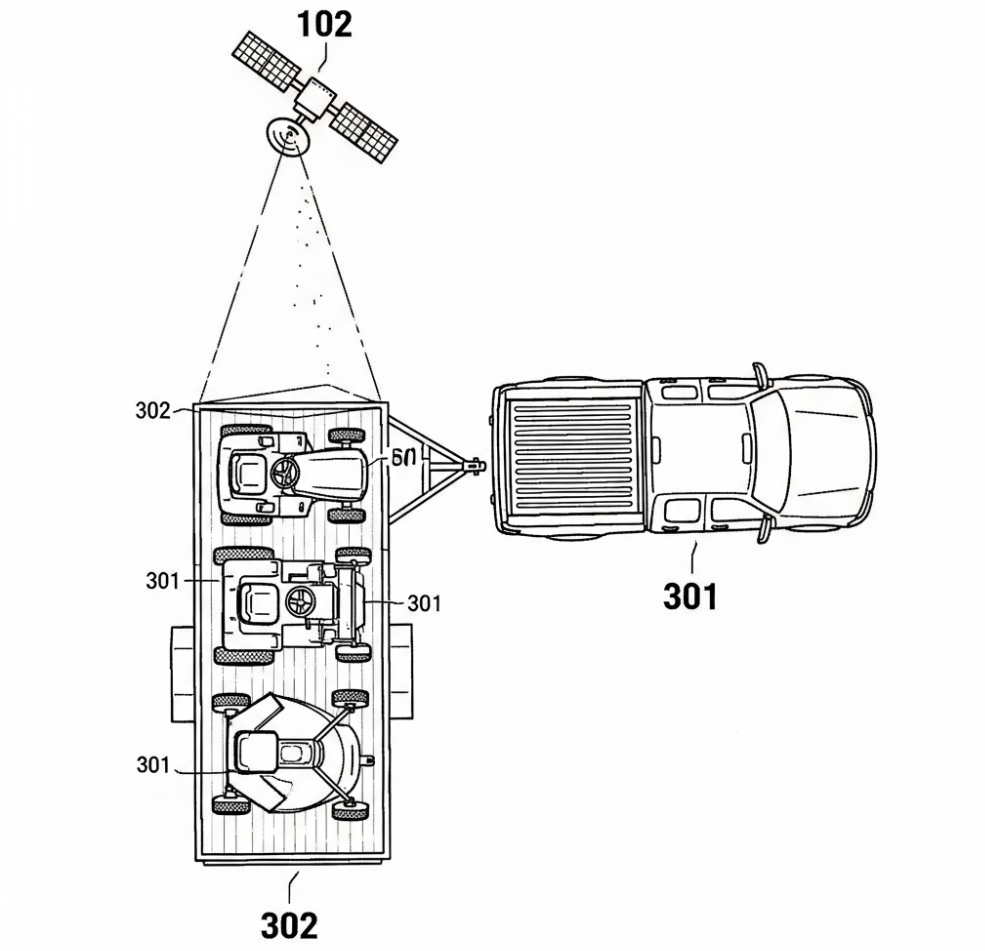


Figure 3: Visual encoder using objects on a trailer.

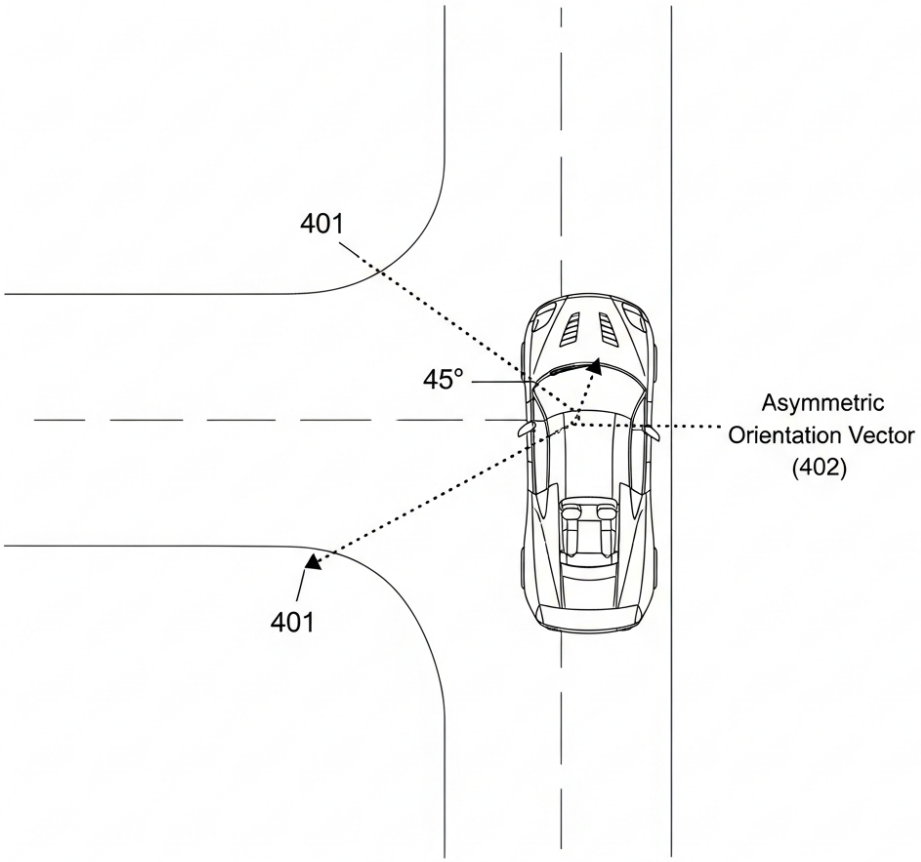


Figure 4: Encoding directionality with a vehicle on a driveway.

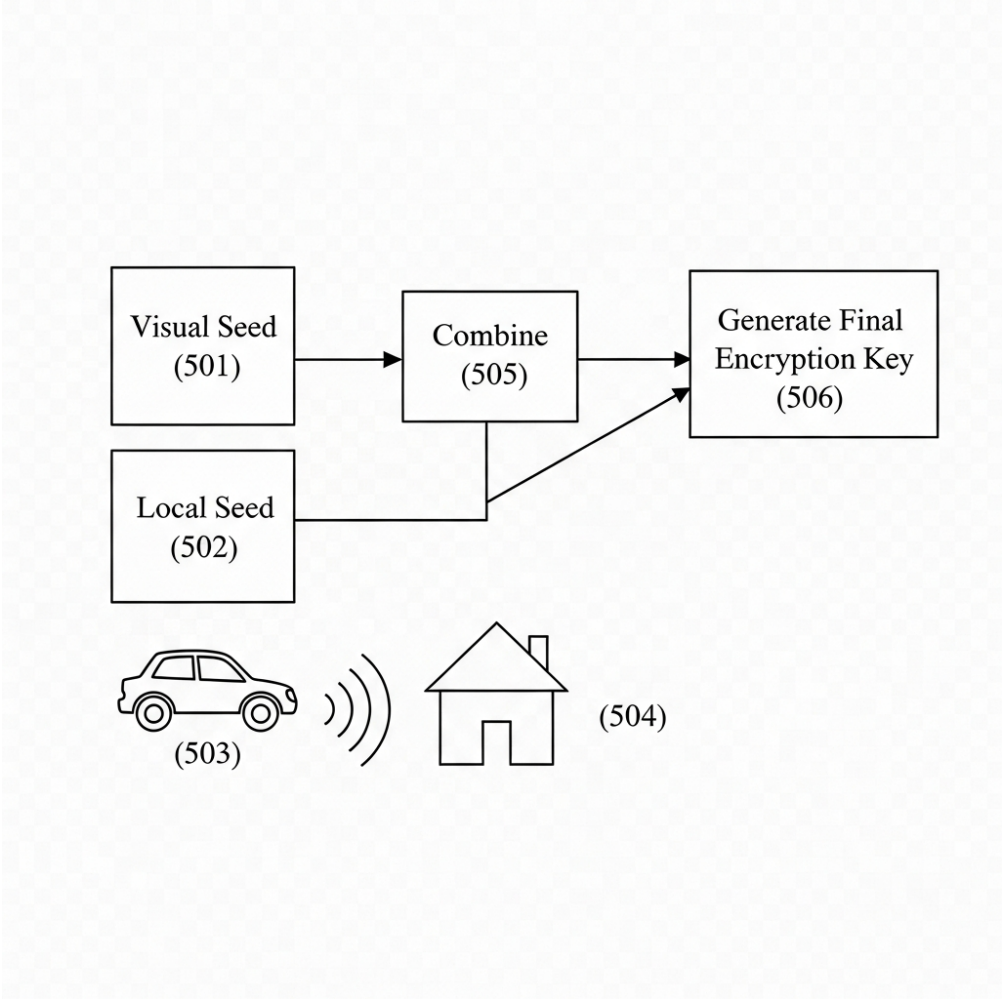


Figure 5: Two-factor key seed transmission method.

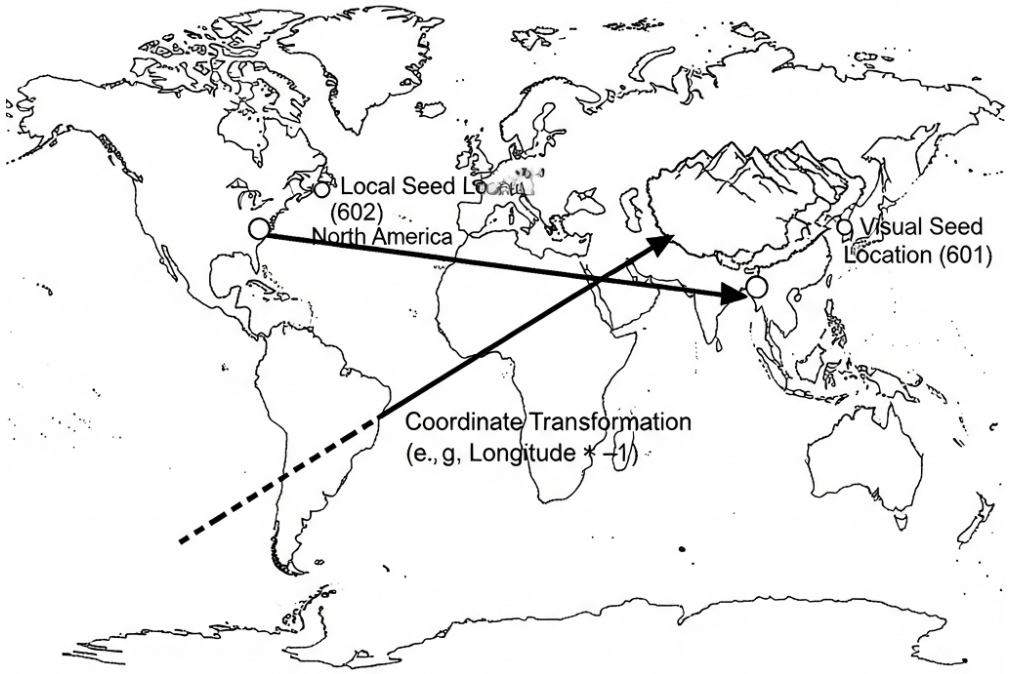


Figure 6: Geo-relocation of a visual key seed.

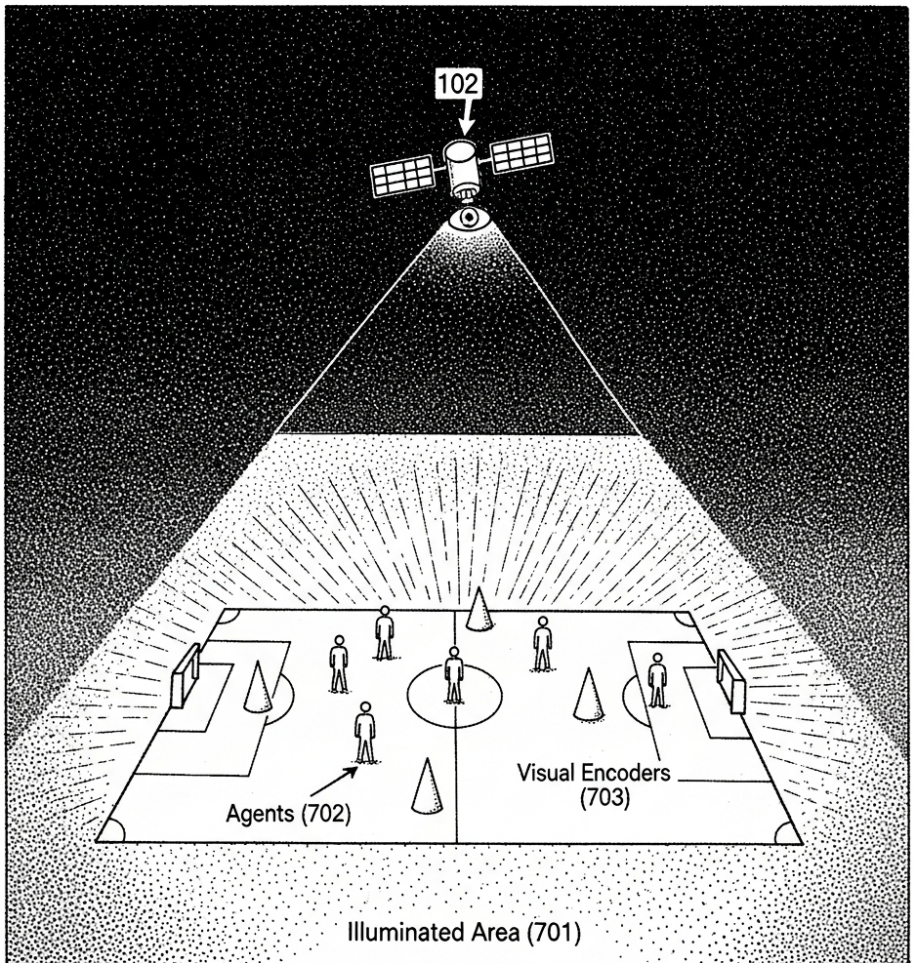


Figure 7: Night-time operation using an illuminated field.